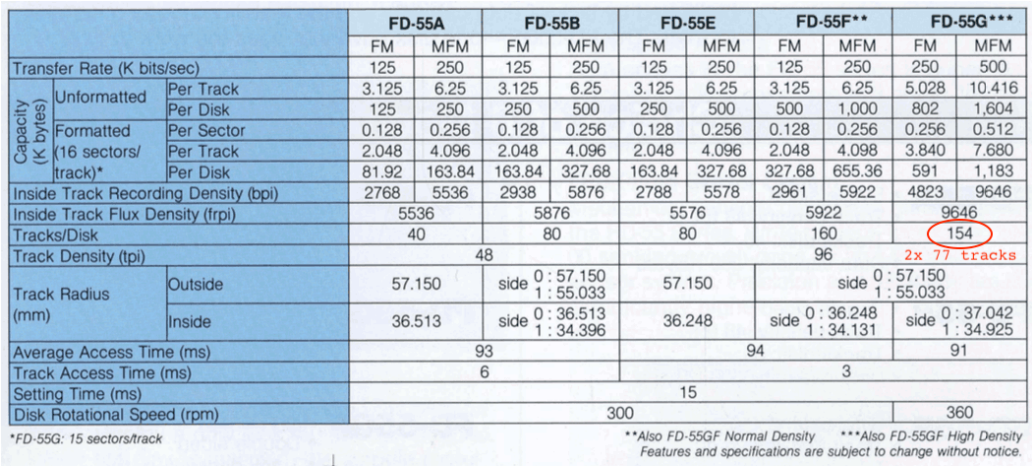
The FD-55 Series

Technical specifications



			FD-55A		FD-55B		FD-55E		FD-55F**		FD-55G***	
			FM	MFM	FM	MFM	FM	MFM	FM	MFM	FM	MFM
Transfer Rate (K bits/sec)			125	250	125	250	125	250	125	250	250	500
Capacity (K bytes)	Unformatted	Per Track	3.125	6.25	3.125	6.25	3.125	6.25	3.125	6.25	5.028	10.416
		Per Disk	125	250	250	500	250	500	500	1,000	802	1,604
	Formatted (16 sectors/ track)*	Per Sector	0.128	0.256	0.128	0.256	0.128	0.256	0.128	0.256	0.256	0.512
		Per Track	2.048	4.096	2.048	4.096	2.048	4.096	2.048	4.098	3.840	7.680
		Per Disk	81.92	163.84	163.84	327.68	163.84	327.68	327.68	655.36	591	1,183
Inside Track Recording Density (bpi)			2768	5536	2938	5876	2788	5578	2961	5922	4823	9646
Inside Track Flux Density (frpi)			5536		5876		5576		5922		9646	
Tracks/Disk			40		80		80		160		154	
Track Density (tpi)			48				96				2x 77 tracks	
Track Radius (mm)		Outside	57.150		side 0 : 57.150 1 : 55.033		57.150		side 0 : 57.150 1 : 55.033			
		Inside	36.513		side 0 : 36.513 1 : 34.396		36.248		side 0 : 36.248 1 : 34.131		side 0 : 37.042 1 : 34.925	
Average Access Time (ms)			93				94				91	
Track Access Time (ms)			6				3					
Setting Time (ms)			15									
Disk Rotational Speed (rpm)			300								360	

*FD-55G: 15 sectors/track

**Also FD-55GF Normal Density ***Also FD-55GF High Density
Features and specifications are subject to change without notice.

TEAC FD55 technical specifications
(Source: Brochure TEAC FD-55 Series)

The TEAC mini flexible disk drive series FD-55 includes in principle three main generations. In the first generation there are the drives A to G. The second generation carries the additional letter V and the third the letter R.

In the first generation there is another differentiation. The very first drives do not yet have LARGE SCALE INTEGRATION (LSI) chips, while the following drives do. For example, my 1st FD-55A-00-U has no LSI chips, while in the brochure from 1985 all drives (A through G) do.

1st	2nd	3rd	specs
A	AV	-/-	SS, DD, 48 tpi, 40 tracks, 250 KByte
B	BV	BR	DS, DD, 48 tpi, 80 tracks, 500 KByte
E	EV	-/-	SS, QD, 96 tpi, 80 tracks, 500 KByte
F	FV	FR	DS, QD, 96 tpi, 160 tracks, 1 MByte
G	GV	GR	DS, HD, 96 tpi, 154 tracks, 1.6 MByte
GF	GFV	GFR	DS, HD, 96 tpi, 160 tracks, 1.6 MByte DS, DD, 48 tpi, 80 tracks, 500 KByte
GS	-/-	-/-	DS, HD, 96 tpi, 160 tracks, 1.6 MByte DS, DD, 48 tpi, 80 tracks, 500 KByte

Type B is the double sided version of type A and type F is the double sided version of type E.

Type E and F are not supported on IBM compatible systems! Basically, however, the MS-DOS operating system already supports quad-density drives!

> *The MS-DOS Enzyclopedia, 1988, p. 865*
>
> *FORMAT [drive] /4 - formats a standard double sided, double density disk (360 KB) on a quad-density disk drive.*
>
> *Note: Each OEM determines which switches will be supported by the FORMAT utility included with the versions of MS-DOS sold with its computers.*

Types A, B, E, F have a disk rotational speed of 300 RPM and type G of 360 RPM. They all are single speed drives.

The FD-55G(GV/GR) is the equivalent to an 8 inch, double sided, double density flexible disk drive. For this reason, the G types have 77 tracks per side, or 154 tracks per disk. They all are single speed drives. However, these drives are also capable of reading/writing 80 tracks.

The FD-55GFR drives are double speed drives. You can „change" between 300 and 360 RPM. In other words, type GFR can read/write 250 or 500 KbitB/s; double/normal or high density. But, beware of the size issue with 48 tpi and 96 tpi r/w heads!

The FD-55GS is a SCSI version.

In contrast to the usual 150 Ohm, all drive types of the 1st generation (A, B, E, F, G) use a terminator with a resistance value of 330 Ohm, see „FD-55 Drive Specification", page 111.

There is a another differentiation, with or without the HEAD LOAD SOLENOID (HLS). The brochure of March 1985 states that the HLS is an optional component. My FD-55B-20-U, for example, does not have an HLS, although it is a first generation drive. The periods at PART NO. (P/N) and SERIAL NO. (S/N) serve only for better readability!

mode	P/N	S/N	LSI	HLS
FD-55A-00-U	1930.708.0-00	0.244.862	no	yes
FD-55A-00-U	1930-711.0-00	1.665.862	yes	yes
FD-55B-01-U	1930.708.0-01	0.400.269	no	yes
FD-55B-01-U	1930.711.0-01	2.229.649	yes	yes
FD-55B-20-U	...	...	yes	no
FD-55E-02-U	...	...	yes	yes

FD-55F-03-U	...	yes	yes
FD-55F-10-U	...	no	yes

In irregular intervals I look at ebay for the TEAC FD-55 and copy and collect the sales pictures posted there (last update: 12/25/2022; different types: 68). If you are interested in pictures send me an email.

- FD-55A
  - FD-55A-00-U
  - ----------
  - FD-55AV-10-U
- FD-55B
  - FD-55B
  - FD-55B-01-U
  - FD-55B-20-U
  - ----------
  - FD-55BV-06-U
  - FD-55BV-11-U
  - FD-55BV-31-U
  - FD-55BV-36-U
  - FD-55BV-53-U
  - FD-55BV-75-U
  - FD-55BV-221-U
  - ----------
  - FD-55BR-100-U
  - FD-55BR-101-U
  - FD-55BR-102-U
  - FD-55BR-104-U
  - FD-55BR-105-U
  - FD-55BR-501-U
  - FD-55BR-506-U
  - FD-55BR-521-U
  - FD-55BR-527-U
  - FD-55BR-528-U
  - FD-55BR-578-U
- FD-55E
  - FD-55E-02-U
  - FD-55EV-12-U
- FD-55F
  - FD-55F-03-U
  - FD-55F-10-U
  - FD-55F-15-U
  - FD-55F-30
  - ----------

    - FD-55FV-03-U
    - FD-55FV-13-U
    - FD-55FV-30-U
    - FD-55FV-31-U
    - FD-55FV-52-U
    - FD-55FV-55-U
    - ----------
    - FD-55FR-511-U
    - FD-55FR-529-U
    - FD-55FR-576-U
- FD-55G
    - G .... not yet seen ¯_(ツ)_/¯
    - ----------
    - FD-55GV-14-U
    - FD-55GV-20-U
    - FD-55GV-33-U
    - ----------
    - GR ... not yet seen ¯_(ツ)_/¯
- FD-55GF
    - FD-55GFV-07-U
    - FD-55GFV-17-U
    - FD-55GFV-56-U
    - FD-55GFV-90-U
    - ----------
    - FD-55GFR-140-U
    - FD-55GFR-141-U
    - FD-55GFR-142-U
    - FD-55GFR-143-U
    - FD-55GFR-145-U
    - FD-55GFR-149-U
    - FD-55GFR-149-U5
    - FD-55GFR-153-U
    - FD-55GFR-159-U
    - FD-55GFR-184-U
    - FD-55GFR-193-U
    - FD-55GFR-197-U
    - FD-55GFR-203-U
    - FD-55GFR-220-U
    - FD-55GFR-232-U
    - FD-55GFR-540-U
    - FD-55GFR-541-U
    - FD-55GFR-570-U
    - FD-55GFR-612-U

- FD-55GFR-620-U
- ----------
- FD-55GFR-7149-U5
- FD-55GFR-7193-U
- FD-55GFR-7393-U

Documents

Brochures

- TEAC FD-55 Series (Mar 1985: A, B, E, F, G, GF)
- TEAC FD-55 GS Series (Feb 1991: GS/SCSI)
- TEAC FD-55 R Series (Nov 1990: BR, FR, GR, GFR)

Specification

- TEAC FD-55 Drive Specification
- TEAC FD-55 Instruction Manual
- TEAC FD-55 Maintenance Manual
- TEAC FD-55 AV / GV Maintenance
- TEAC FD-55 BV Drive Specification
- TEAC FD-55 BR Drive Specification
- TEAC FD-55 E / F Drive Specification
- TEAC FD-55 FV Drive Specification
- TEAC FD-55 Jumper Settings

Some drives in the FD-55 series

--> FD-55A: The Single
--> FD-55E: The Odd one
--> FD-55F: The Specialist / *Runing under DOS*

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